

$$= \frac{nm}{2} \quad \square$$

Case 2: q is False, but r is True. $\therefore n$ is even

We cover the checkerboard using the following strategy:

Row 1: Column 1, 2 $\rightarrow$ Dominoe 1 Column 3, 4 $\rightarrow$ Dominoe 2, ...,
 Column $j, j+1 \rightarrow$ Dominoe $(\frac{j+1}{2})$, ..., Column $n-1, n \rightarrow$ Dominoe

$n/2$.
Row 2: Column 1, 2 $\rightarrow$ Dominoe $n/2+1$ Column 3, 4 $\rightarrow$ Dominoe $n/2+2$, ...,
 Column $j, j+1 \rightarrow$ Dominoe $(\frac{j+1}{2}) + \frac{n}{2}$, ..., Column $n-1, n \rightarrow$ Dominoe

$$n/2 + n/2 = 2n/2$$

Row i : Column 1, 2 $\rightarrow$ Dominoe $(i-1)\frac{n}{2}+1$ Column 3, 4 $\rightarrow$ Dominoe
 $(i-1)\frac{n}{2}+2$, ..., Column $j, j+1 \rightarrow$ Dominoe $(i-1)\frac{n}{2} + (\frac{j+1}{2})$, ...,
 Column $n-1, n \rightarrow$ Dominoe $(i-1)\frac{n}{2} + \frac{n}{2} = \frac{in}{2}$

Row m : Column 1, 2 $\rightarrow$ Dominoe $(m-1)\frac{n}{2}+1$ Column 3, 4 $\rightarrow$ Dominoe
 $(m-1)\frac{n}{2}+2$, ..., Column $j, j+1 \rightarrow$ Dominoe $(m-1)\frac{n}{2} + (\frac{j+1}{2})$, ...,
 Column $n-1, n \rightarrow$ Dominoe $(m-1)\frac{n}{2} + \frac{n}{2} = \frac{mn}{2} \quad \square$

46. Prove or disprove that you can use dominoes to tile a 5×5 checkerboard with three corners removed.

Solution: WLOG, let the uppermost left square be white. As an evidence from the 5×5 checkerboard shown, there are 13 white squares and 12 black squares.

W	B	W	B	W
B	W	B	W	B
W	B	W	B	W
B	W	B	W	B
W	B	W	B	W